

NAME: _____

PERIOD: _____

DATE: _____

Can Chance Make a Streak?

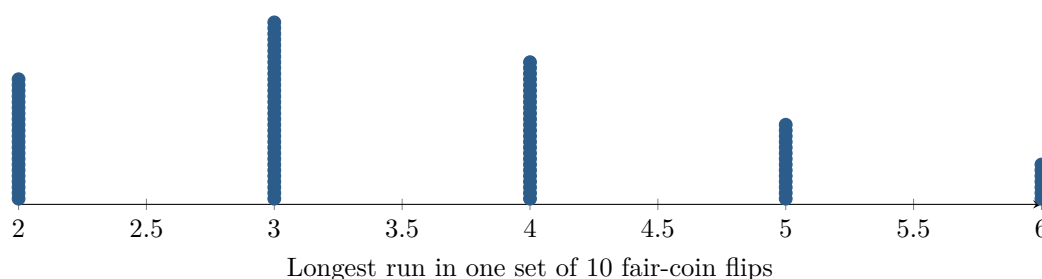
AP Statistics · Unit 2 · Topic 2.3 · B: 2026-10-19 · E: 2026-11-02

[STAR] TODAY'S PROBLEM

Suppose a coin advertised as fair is flipped 10 times and gives HHHHHHTTTT. To investigate what chance alone could produce, a class simulates 100 sets of 10 independent fair-coin flips. For each set, the class records the longest run of matching outcomes; the dotplot shows the results.

Claim A: “The coin is rigged—six heads in a row cannot be chance.”

Claim B: “Streaks like this can happen by chance; simulation can tell us how unusual this one is.”



FIRST TAKE — before the video (5 min)

Which claim seems better supported right now? Cite one detail from the sequence or dotplot.

[BOOK] MODEL THE PATTERN, NOT THE STORY (keep on the page)

Random processes can produce visible patterns, including streaks. To judge a pattern under a proposed chance model, make each simulation trial match the original process, choose a statistic that measures how extreme the pattern is, and count results **at least as extreme** as the observation. Repeat many independent trials; the successful-trial proportion estimates how often chance alone would produce such a result. An unusual result is evidence against a model, not proof of what caused the departure.

Finish after the video:

DOK 1 (a) State the longest run in the observed sequence and the most common longest-run value among the simulated sets.

DOK 2 (b) Use the dotplot to estimate the probability that 10 fair-coin flips have a longest run at least as long as the observed run. Interpret the estimate in context.

DOK 3 (c) Which claim is better supported by this evidence? Cite the simulation result that settles your choice. Then describe a larger simulation that would more precisely settle the disagreement: state what one trial is, what to record and count, how many trials to use, and what result would make you doubt that the coin is fair.

Frame: Claim ____ is better supported because _____.

Frame: One trial is _____; I would record _____ and doubt fairness if _____.

EXIT — turn this in

One thing the video changed about my first take: _____